Dataset Documentation

1. Dataset Overview

The TexCycle AI platform uses structured textile-material data and image data to support its AI components. The datasets support fabric classification, defect analysis, sustainability prediction, commercial analysis, and marketplace forecasting.

The dataset development process uses information gathered through textile industry visits and global and Sri Lankan industry reports.

2. Dataset Structure

Sustainability Intelligence Dataset

Input features

Feature	Description
fabric_type	Textile material category
weight_kg	Material weight in kilograms
health_score	Material health condition score
repairability	Material repairability information
condition	Current material condition
district	District associated with the material
province	Province associated with the material
industry_type	Type of textile-related industry

Target Variables

The regression models generate predictions for:

- circularity_score
- recyclability_score
- carbon_reduction_percent

- water_reduction_percent
- co2_saved_kg
- water_saved_liters

A separate classification model predicts the sustainability grade.

3. Data Preprocessing

The Sustainability Intelligence pipeline uses the input features required by the trained models. Model outputs are constrained to defined ranges to maintain valid prediction values. Numerical outputs are rounded to two decimal places.

The implementation also derives additional indicators using predefined formulas. These include energy savings, landfill diversion, SDG impact score, and economic impact score.

4. Machine Learning Models

The Sustainability Intelligence component uses:

- **XGBoost Regressors:** Predict numerical sustainability indicators.
- **XGBoost Classifier:** Predicts the sustainability grade.
- **Label Encoding:** Converts grade labels into the format required by the classification model.
- **Joblib:** Supports loading serialized model-related objects.

5. Model Output Processing

The model outputs are processed using the following limits:

- Circularity, recyclability, carbon reduction, and water reduction: 0–100.
- CO_2 savings: 0 to five times the material weight in kilograms.
- Water savings: 0 to 300 litres per kilogram of material.

Energy savings, landfill diversion, SDG impact, and economic impact are calculated using the formulas implemented in the model pipeline.

6. Model Artifacts

The implementation loads trained model artifacts, including regression models and a grade classification model, from JSON files. Label-encoding information is also used for grade prediction.

7. Data Limitations

The predictions represent model-based estimates rather than direct environmental measurements. Further validation using independent textile-market data and real transaction records is needed to evaluate generalisation and performance in the Sri Lankan marketplace.

8. Reproducibility

For reproducible results, the dataset version, feature definitions, preprocessing steps, model artifacts, training configuration, and evaluation metrics should be maintained alongside the implementation.